Newbear 77-68 : 1977
Owned by Newbury Laboratories:
Motorola 6800 8-bit processor

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SIMON (**S**oftware **I**nterrupt **MON**itor program) 1KB
DEMON = SIMON+1KB extension

In 1979 one of my brothers (Mike) worked as tech. support in Newbear under Tim Moore who designed the Newbear 77-68. My other brother (Dave) built one as did a workmate of mine at AERE Harwell (Mike Davies, sadly deceased in 2021). Mike Davies***** and I had a DEC PDP-11 at work at the time and so by staying on after work hours Mike.D wrote a cross-assembler for the 6800, which I used to enter code, such as the MINIMON monitor source so I could modify it easliy to make whatever changes Dave and Mike.D wanted, removing some unwanted code and adding support for a 2nd ACIA serial interface.

At that time, I had an ambition to write my own monitor which would support the use of Software Interrupts to perform functions, such as print a character to the serial port, rather than rely on application programs knowing the exact locations of these in the monitior and thus having to call then using a specific address, which meant that any changes to the monitor which moved code meant having to change all applications which called it.

So, instead of something like:-

```
JSR  <address-of-print-char-routine>
```

The shorter and simpler, SWI followed by a function code is used:-

```
SWI                 Sofware interrupt to monitor
FCB     $01         Function.1 = Print Byte (from Reg.A)
```

I had test versions of the software interrupt handling code, along with some functions take from the MINIMON and MIKBUG monitors incorporated into a program called DEMON. Due to changes of job and general life getting in the way, the unfinished DEMON was put aside around 1980. Dave continued using his 77-68 until the mid 1980s, when if was put into storage. A bit over 25 years later, Dave recovered his 77-68 from the loft and over the next few years managed to build Mike.Ds old 77-68, which he had been given, into a crate and also make a 3rd 77-68 system based on a motherboard Mike.D had somehow got hold of. These were know as `Twit`**, `Twit.MkII` and `Twit.NT`.

***** Mike Davies was kown by most people as `Big Mike` **-** a term originally used to differentiate him from my brother Mike.

** `Twit` is an old Goon Show reference to and `Electric Twit` and `NT` is `New Technology` as this one had some slightly newer chips making up the RAM/ROM and I/O board.

I finally got a fully working DEMON monitor working in 2018, mostly debugging on `Twit.NT`, which Dave had passed to me by that time.

The DEMON monitor takes up 2KB of ROM (or RAM) plus it uses the motherboard 256byte area for variables and its stack.

SIMON is based on DEMON, but effectively split into two 1KB parts - the main Software Interrupt handling part, I/O routines etc in the the top 1KB (FC00 to FFFF) and containing the **M**odify, **L**oad and **G**o commands (only). (SIMON)

The second, optional Extras, part can potentially be anywhere (and not limited to 2KB) and contains other commands, such as **D**ump, **B**lock move, **P**unch a help (**?**) command and some memory test functions. (SIMON + this extension = DEMON)

The main component of SIMON checks for its own built-in commands, if it does not understand a command, then it checks for the Extras component, which begins with a `magic` 2 byte number at F800 (currently the address for the Extras can only be changed by changing an EQU statement and recompiling SIMON).

<< *more notest to be added* >>